12 Assimilating Seizures

As meteorologists have long known, the atmosphere exhibits no periodicities of the kind that enable one to predict the weather in the same way one predicts the tides. No simple set of causal relationships can be found which relate the state of the atmosphere at one instant of time to its state at another. It was this realization that led V. Bjerknes in 1904 to define the problem of prognosis as nothing less than the integration of the equations of motion of the atmosphere. But it remained for Richardson to suggest in 1922 the practical means for the solution of this problem. He proposed to integrate the equations of motion numerically and showed exactly how this might be done. That the actual forecast used to test his method was unsuccessful was in no sense a measure of the value of his work. In retrospect, it becomes obvious that the inadequacies of observation alone would have doomed any attempt however well conceived, a circumstance of which Richardson was aware. The real value of his work lay in the fact that it crystallized once and for all the essential problems that would have to be faced by future workers in the field and that it laid down a thorough ground-work for their solution.
—Jule Charney, 1951 [Cha51]

12.1 Introduction

By the turn of the twentieth century, Vilhelm Bjerknes had proposed, in 1904 [Bje04], that weather prediction required the use of the fundamental equations of motion of the atmosphere. The full equations of such motion [Kal03] were intractable to integrating by hand, there was no regional or global network of weather sensors, and digital computers would not be invented for decades.

When these issues began to be addressed, the first results were worthless. The introduction of simplified dynamical models that retained the most important of the physical dynamics was a critical development [Kal03]. The turning point came when integrating simplified models gave "first approximations that bore a recognizable resemblance to the actual motions" (Jule Charney, 1951 [Cha51] discussing his historic 1950 demonstration with John von Neumann that integrating a fundamental model made a meaningful 1-day weather forecast feasible [CFVN50]).

It is the author's conjecture that at the time of the writing of this book, neuroscience shares similarities with Charney's predicament nearly six decades before. This chapter will be the first description of our efforts at integrating the equations of motion of biophysically

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realistic models of neurons, using data assimilated from experiments. Our findings will indeed bare a recognizable resemblance to the actual neuronal dynamics. I have placed this chapter at the end of this book, but it is just the beginning.

12.2 Hodgkin-Huxley Revisited

Neuronal activity is almost always measured by observing just a single variable such as voltage (or calcium using optical techniques). Such measurements are always uncertain, due to a combination of noise in neurons and amplifiers, as well as uncertainties in recording equipment such as electrode access resistance and capacitance. And such measurements only reflect one of the many variables that are important in the dynamics of an electrically active neuron.

In chapter 3 we introduced the foundational Hodgkin-Huxley equations [HH52a, HH52b, HH52c, HH52d]. In subsequent chapters, we demonstrated that reduced models of Hodgkin-Huxley type dynamics could be employed in a control framework (chapters 5 and 10). But what about using the original equations? The distinction is important—the full Hodgkin-Huxley formalism includes the actual biophysical variables such as voltage and ionic currents.[1]

Of course, Hodgkin and Huxley had already placed their system in a control framework. Their goal was to use feedback to control the transmembrane voltage to a given constant command potential. The currents required to maintain the voltage in response to transient voltage changes are the currents that entered into their original equations. More recently, *dynamic clamp* techniques have introduced the manipulation of a conductance by adjusting the amount of current injected into a neuron while simultaneously measuring the membrane potential [GM06, PAM04]. This has typically been done by using the conductance equation of a particular current, measuring voltage across the membrane, and reconstructing the current required to add or subtract the particular current of interest. The reader can likely see where this line of thought is going to lead, but we need to address the original Hodgkin-Huxley equations first.

The transmembrane potential V of a single neuron, normalized to zero in the resting state, is modeled with the Hodgkin-Huxley equations (3.9). We can write these equations in compact form as

$$
\begin{aligned}
V &= e + \frac{1}{C_m}\int_0^T (I_K + I_{Na} + I_L + I_{stim})dt,\\
I_K &= -g_K n^4 (V - V_K), \quad I_{Na} = -g_{Na} m^3 h (V - V_{Na}),\\
I_L &= -g_L (V - V_L),\ dq/dt = \alpha_q (1-q) - \beta_q q,\ q = m, n, h,
\end{aligned}
\tag{12.1}
$$

1. The author prematurely slipped into the chapter on Parkinson's disease (chapter 10) a Hodgkin-Huxley model in figure 10.19, without offering an adequate discussion for such modeling in control frameworks.

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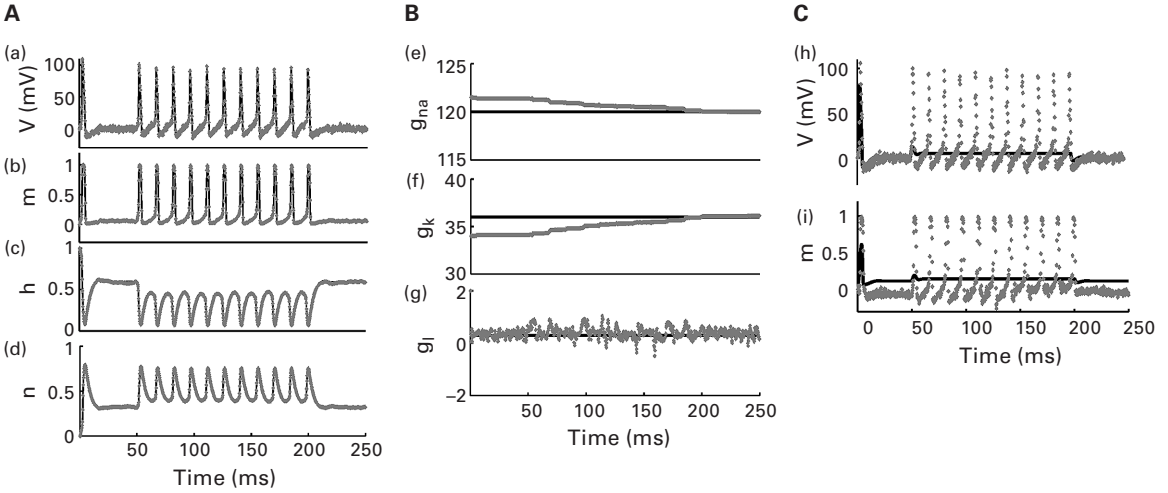


Figure 12.1
Reconstruction of full Hodgkin-Huxley dynamics through noisy voltage measurements alone. A(a–g) tracks the dynamics of the HH neuron (true values from model are shown in the black line, and uncontrolled estimates are gray symbols and lines). Fixed reversals of -12 mV and 10.6 mV (relative to 0 mV resting potential), respectively, are used for K^+ and leak currents (external potassium concentration $[K]_o$ fixed), and a constant stimulus I_{stim}=10 μA/cm^2 is applied to the neuron for $50 \leq t \leq 200$ msec. In B (h, i) inaccurate model gives accurate estimates—same as a, b except constant α_m is used ($\alpha_m = 0.5$ black, α_m estimated gray). Reproduced from [US09] with permission.

where e is the V measurement noise;[2] n^4 and m^3h represent gating variables for potassium, I_K; and sodium, I_{Na}, currents; and I_L is the leak current. The rate equations for the gating variables, written with the modern convention for voltage sign, are $\alpha_n = (0.1 - 0.01V)/(\exp(1 - 0.1V) - 1)$, $\beta_n = 0.125\exp(-V/80)$, $\alpha_m = (2.5 - 0.1V)/(\exp(2.5 - 0.1V) - 1)$, $\beta_m = 4\exp(-V/18)$, $\alpha_h = 0.07\exp(-V/20)$, and $\beta_h = 1/(\exp(3 - 0.1V) + 1)$.

In contrast with our previous employment of reduced models [VTK04, SS08], we here seek to reconstruct the full Hodgkin-Huxley ionic dynamics, including gating variables of the channels and various conductance parameters, using noisy voltage measurements. Although there are more sophisticated methods to explicitly account for model inadequacy (chapter 8), here we minimize model inadequacy by carefully choosing the process noise added to the voltage estimate.

Figure 12.1 shows the voltage measurements, with the estimated gating variables m, h, and n in panels a–d, respectively. We also tracked the three current conductances—g_{Na}, $g_k K$, and g_L—shown in panels e–g, respectively.

2. I am going to substitute e for r in this chapter, in order to reduce the chance that a reader will missread r for "resistance."

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The reconstructions shown in figure 12.1 of the full Hodgkin-Huxley equations are far better than what seems reasonable. Granted, this is the assimilation of data, contaminated with noise, where the model producing the data and the model doing the assimilation are the same. But this is far better than what one would see using, for instance, the Wilson-Cowan equations [SS09] (chapter 6). It is very likely that symmetries are important.[3]

In the first of Hodgkin and Huxley's four 1952 papers [HH52a], they assumed "only that the chance that any individual ion will cross the membrane in a specified interval of time is independent of the other ions which are present." This *independence principal* was borne out by experiments (chapter 3), and it is reflected in the Hodgkin-Huxley equations. Each major ion, Na and K, was indeed shown to travel through separate channels in work that followed Hodgkin and Huxley's original curve fits. Of considerable help to us is the fact that these two major currents are also operating on different time scales. But this does not completely account for the absurdly high-quality fits in figure 12.1. Even the parameters of maximal conductance—g_{Na}, g_K, and g_L—converge rapidly to their true values (the process noises, Q, used were 1.5, 0.5, and 0.5 for g_{Na}, g_K, and g_L, respectively).

Let's formally make this an inadequate model. By replacing the dynamics of I_{Na} turn-on with a constant, we destroy the ability of the equations to spontaneously spike. We do this by substituting the actual $\alpha_m(V)$

$$\alpha_m(V) = (2.5 - 0.1V)/(\exp(2.5 - 0.1V) - 1)$$

with the trivial

$$\alpha_m(V) = a \text{ (a constant)}$$

This is not the first time that we have employed a constant to track a dynamical parameter. A similar example of such use was the trivial dynamics in the reconstruction of the unknown signal input in the Fitzhugh-Nagumo equations in chapter 4. Recall that the Kalman filter will continually compare the value of such a floating constant with the reconstruction, and fit a compromise value using Bayes's rule. Each iteration propagates this new value forward as the next value of a. Trivial dynamics such as these are not predictive. They assume that the parametric values are just not very different from the previous value. For parameters that are relatively slowly changing compared with faster-moving variables, such trivial dynamical assumptions can be very good. In this particular case, we have done this with one of the fastest changing variables in the set of equations, α_m. Intriguingly, although the reconstruction is off by a modest amount (20 to 30% root-mean-square, time-step dependent, of the full functional form of α_m), the reconstruction is sufficiently good to track the spiking as shown in figure 12.1B. Kalman filters are robust for this type of tracking even in the face of potentially crippling model inadequacy. Kalman filtering constantly reestimates the

3. An important discussion of symbolic observability can be found in Letellier and Aguirre [LA09] and references therein.

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trade-off between model accuracy and measurements, expressed in the filter gain function [Kal60, Sim06a].

Hodgkin and Huxley performed their experiments on the giant axon of the squid in a perfusion chamber where they assumed that the ion concentrations outside of their nerve membrane were constant. This is not unreasonable given that they stripped out a nearly bare axon, and immersed it in an ionic bath of, for all practical purposes, infinite extent. Hodgkin and colleagues would later revisit the important issue of how ions change in the extracellular space, especially potassium, when "No attempt was made to remove the layer of connective tissue (about 20 μm in thickness) which clings tightly to the axon" [FH56]. Even a single action potential leaves a measurable and physiologically important accumulation of $[K^+]_o$ outside of the neuronal membrane in its wake. In [FH56] the increase with each action potential was about 1.6 mM, and it dissipated with a time constant of 30 to 100 msec. There is a rich subsequent history of the consequences of such potassium changes in the extracellular space of brains [Som04].

So let's force $[K^+]_o$ to vary substantially. The potassium and leak reversal potentials will be updated based on the instantaneous $[K]_o$ using the Nernst equation

$$V_k = 70 + 26.64 \ln([K]_o / [K]_i) \tag{12.2}$$

and the Goldman-Hodgkin-Katz equation [Gol43, HK49]

$$V_l = 70 + 26.64 \ln\left(\frac{[K]_o + 0.085[Na]_o + 0.1[Cl]_i}{[K]_i + 0.085[Na]_i + 0.1[Cl]_o}\right) \tag{12.3}$$

In figure 12.2, we show the results of forcing the Hodgkin-Huxley equations through cyclically modulating the $[K^+]_o$ between 5 and 10 mM. Note that the neuron spikes on the

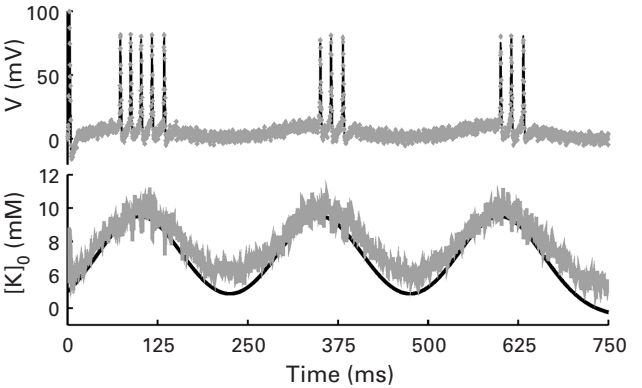


Figure 12.2
Driving Hodgkin-Huxley equations through variable external potassium concentration, $[K^+]_o$. Only noisy voltage, V, is measured. Reconstructed voltage and $[K^+]_o$ are in gray, with underlying true values in black. Reproduced from [US09] with permission.

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peaks of this variation in $[K^+]_o$. We now measure noisy voltage. Again, reconstruction of voltage and the external potassium $[K^+]_o$, as well as the remaining variables and parameters from figure 12.1 (not shown), are well estimated.

12.3 The Dynamics of Potassium

We have learned much about the dynamics of potassium flow in the brain in the years following the study of isolated squid axons. In general, the intrinsic excitability of neuronal networks depends on the reversal potentials for ion species. During neuronal activity, both the extracellular potassium and intracellular sodium concentrations ($[Na^+]_i$) increase [MFP74, HLG77, RRS00, AMM02]. Glia help to reestablish the normal ion concentrations, but require time to do so. Consequently, neuronal excitability is transiently modulated in a competing fashion: the local increase in $[K^+]_o$ raises the potassium reversal potential, increasing excitability, while the increase in $[Na^+]_i$ leads to a lower sodium reversal potential and thus less ability to drive sodium into the cell. The relatively small extracellular space and weak sodium conductances at normal resting potential can cause the transient changes in $[K^+]_o$ to have a greater effect over neuronal behavior than the changes in $[Na^+]_i$, and the overall increase in excitability can cause spontaneous neuronal activity [McB94, RLJ85, TD88]. Nevertheless, the competing influences of these two ions on excitability also sets up neurons and their networks to oscillate in their excitability, as we will shortly demonstrate.

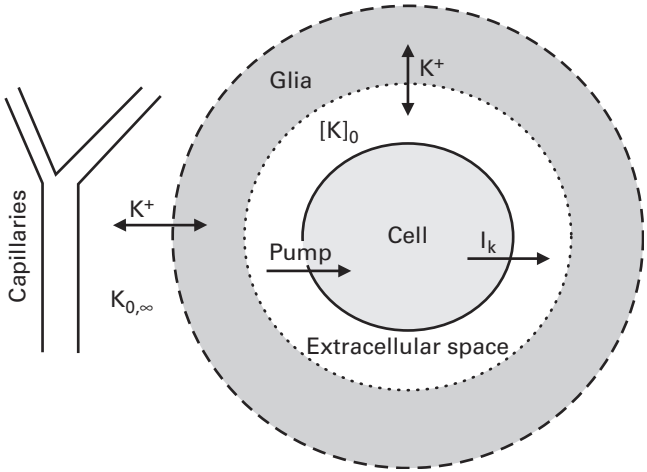


Figure 12.3
A schematic of the model dynamics: the cell releases K^+ through I_k, which is either pumped back into the cell through K^+-Na^+ exchange pumps or buffered by the glia from the extracellular region. K^+ can also diffuse to the nearby reservoir (capillaries in tissue or bath solution in vitro). Reproduced from [US09] with permission.

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A rich literature on the interrelated dynamics of ion concentration and neurons is evolving [BTSS04, KWS00, KWS02, Som04, PD06, KWS07, FTSB08, SKW08, CUZ+09, UCJBS09]. We will employ a formulation of these dynamics suitable for our purposes, as more fully discussed in [CUZ+09, UCJBS09].

A schematic of the spatial relationship between the spaces involved in our ion dynamics is illustrated in figure 12.3.

J. Robert Cressman and colleagues [CUZ+09] let the extracellular potassium dynamics, $[K]_o$, be represented in a model based on I_K, activity of the pump exchanging K^+ and Na^+, I_{pump}, diffusion of potassium to the microenvironment, I_{diff}, and glial buffering, I_{glia}.

$$[K]_o = e_k + \int_0^T (0.165 I_k - 2\beta I_{pump} - I_{diff} - I_{glia})dt, \tag{12.4}$$

where e_k is the $[K]_o$ measurement noise.

The equations for I_{pump}, I_{diff}, and I_{glia} are, from [CUZ+09]

$$I_{pump} = I_{max}\left(\frac{1}{1+exp((25-[Na]_i)/3)}\right)\left(\frac{1}{1+exp(8-[K]_o)}\right) \tag{12.5}$$

$$I_{diff} = \kappa([K]_o - k_{o,\infty}) \tag{12.6}$$

$$I_{glia} = \frac{G_{glia}}{1+exp((18-[K]_o)/2.5)} \tag{12.7}$$

where the Na^+-K^+ pump is modeled as a product of sigmoidal functions, I_{max} is the pump strength under normal conditions, and $[Na]_i$ is the intracellular sodium concentration. Additional parameters are given in the footnote.[4] Each sigmoidal term saturates for high values of internal sodium and external potassium, respectively. More biophysically realistic models of pumps, such as those in Lauger [Lau91], produced substantially similar results. $k_{o,\infty}$ in the diffusion equation is the potassium concentration in the nearby reservoir. Physiologically, this would correspond to either the bath solution in a slice preparation or the vasculature in the intact brain (noting that $[K]_o$ is kept below the plasma level by transendothelial transport). Both active and passive K^+ uptake into glia is incorporated into a simplified single sigmoidal response function that depends on extracellular K^+ concentration, with G_{glia} representing the maximum buffering strength. A similar but more physiological approach was used in Kager and colleagues [KWS07]. Two factors allow the

4. Membrane capacitance $C_m = 1\mu$F/cm^2, maximum conductances (mS/cm^2) are $g_{na} = 120$, $g_k = 36$, $g_l = 0.3$, sodium reversal $V_{na} = 115$ mV, ion concentrations (mM) are intracellular $[K]_i = 130$, extracellular $[Na]_o = 130$, intracellular $[Na]_i = 20$, extracellular $[Cl]_o = 130$, intracellular $[Cl]_i = 8$ [BTSS04], maximum pump flux $I_{max} = 0.1$mM/sec, and glial buffer strength $G_{glia} = 100$ mM/sec. The diffusion coefficient of $[K]_o$ to the nearby reservoir κ, from Fick's law, is $\kappa = 2D/\Delta x^2 = 4.0$ sec^{-1}, where $D = 400 \times 10^{-6}$cm^2/sec is K^+ diffusion constant in neocortex [FPP76] and $\Delta x \approx 15\mu$m for brain reflecting the average distance between capillaries [Sch44a]. The factor 0.165 mM·cm^2/μcoul in eq.(12.7) converts ionic current to concentration rate of change and is calculated using $\beta A/FV$ [CUZ+09], where A, V and F represent cell area, volume and Faraday constant, respectively.

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glia to provide a nearly insatiable buffer for the extracellular space. The first is the large size of the glial network. Second, the glial end-feet surround the pericapillary space, which, through interaction with arteriole walls, can effect blood flow; this in turn increases the buffering capability of the glia [KWBT71, MEW82, PN87].

To complete the description of the potassium concentration dynamics, we assume that the flow of Na^+ into the cell is compensated by flow of K^+ out of the cell. Then $[K^+]_i$ can be approximated by

$$[K]_i = 140.0\,mM + (18.0\,mM - [Na]_i)$$

where 140.0 mM and 18.0 mM reflect the normal resting $[K^+]_i$ and $[Na^+]_i$ respectively. The limitations of this approximation are addressed in Cressman and colleagues [CUZ+09].

The intra- and extracellular sodium concentration dynamics are modeled by

$$\frac{d[Na]_i}{dt} = 0.33\frac{I_{Na}}{\beta} - 3I_{pump}$$

and

$$[Na]_o = 144.0mM - \beta([Na]_i - 18.0mM)$$

where we assume that the total amount of sodium is conserved. Here, 144.0 mM is the sodium concentration outside the cell under normal resting conditions for a mammalian neuron.

We will let I_K flow from a spherical cell with a radius of 13 μm for now. We set K^+ in the nearby infinite reservoir $[K^+]_\infty = 4.0$ mM, and intra- to extracellular volume ratio $\beta = 7$ [MTD90]. *Note that this reservoir is not the extracellular space, where potassium dynamics are quickly changing.*

As with Parkinson's disease dynamics (chapter 10), it is instructive to reduce this complex ion model so that we can examine the essence of the excitability changes and study the bifurcation structure of qualitative changes in the dynamics. We formulated this reduction by eliminating the fast-time-scale spiking behavior in favor of the slower ion concentration dynamics. This was accomplished by replacing the entire Hodgkin-Huxley mechanism with empirical fits to time-averaged ion currents. Using the membrane conductances from the full model, we fixed the internal and external sodium and potassium concentration ratios and allowed the model cell to attain its asymptotic dynamical state, which was either a resting state or a spiking state. Then, the sodium and potassium membrane currents were time averaged over 1 sec. These data were fit to products of sigmoidal functions of the sodium and potassium concentration ratios, resulting in the (infinite-time) functions

$$I_{Na\infty}([Na]_i/[Na]_o, [K]_o/[K]_i)$$

and

$$I_{K\infty}([Na]_i/[Na]_o, [K]_o/[K]_i)$$

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The interested reader can find full details in the appendix of Cressman and colleagues [CUZ+09]. $I_{Na\infty}$ is nearly identical to $I_{K\infty}$, differing significantly only near normal resting concentration ratios, due to differences in the sodium and potassium leak currents. These currents replace the original I_{Na} and I_K in the full model described earlier. We examine the estimated $[K^+]_o$ and $[Na^+]_o$ produced by the full and reduced model during episodic seizure like activity when the $[K^+]_\infty$ of the distant reservoir is increased to two times normal levels, as shown in figure 12.4.

I have dragged the poor reader through this morass of ionic notation to be able to show the following. Using the reduced model, we can analytically examine the interrelationships of varying external potassium concentration in the distant reservoir, $[K^+]_\infty$, ion pump strength, I_{pump}, diffusion strength, I_{diff}, and glial buffering strength, I_{glia}, as shown in figure 12.5. We see that as is well known experimentally [RLJ85], increasing $[K^+]_\infty$ to near twice normal results in oscillations in extracellular potassium (figure 12.5A), which will cause the excitability of the neurons to oscillate as illustrated in figure 12.2. At higher concentrations of reservoir potassium, a stable level of extracelluar potassium reemerges, which corresponds to a cessation of spiking during a depolarization block (the far right

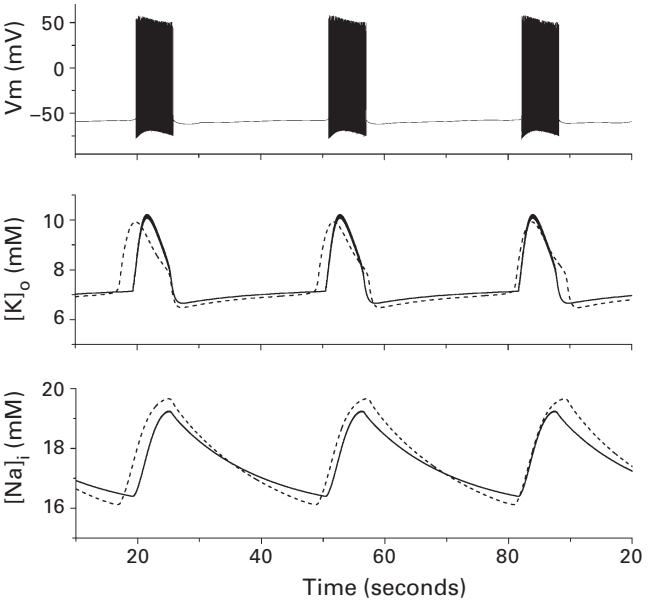


Figure 12.4
Comparison of full (solid line) versus reduced (dashed line) model of ion dynamics when potassium in the distant reservoir $[K^+]_\infty$ is doubled over normal. Episodic periods of high firing activity, shown in the upper tracing, last many seconds, similarly to intermittent seizures. These periods of high activity are reflected in the substantial ion shifts shown. The reduced model is a good qualitative approximation to the full ion dynamics occurring during these events. Reproduced from [CUZ+09] with permission.

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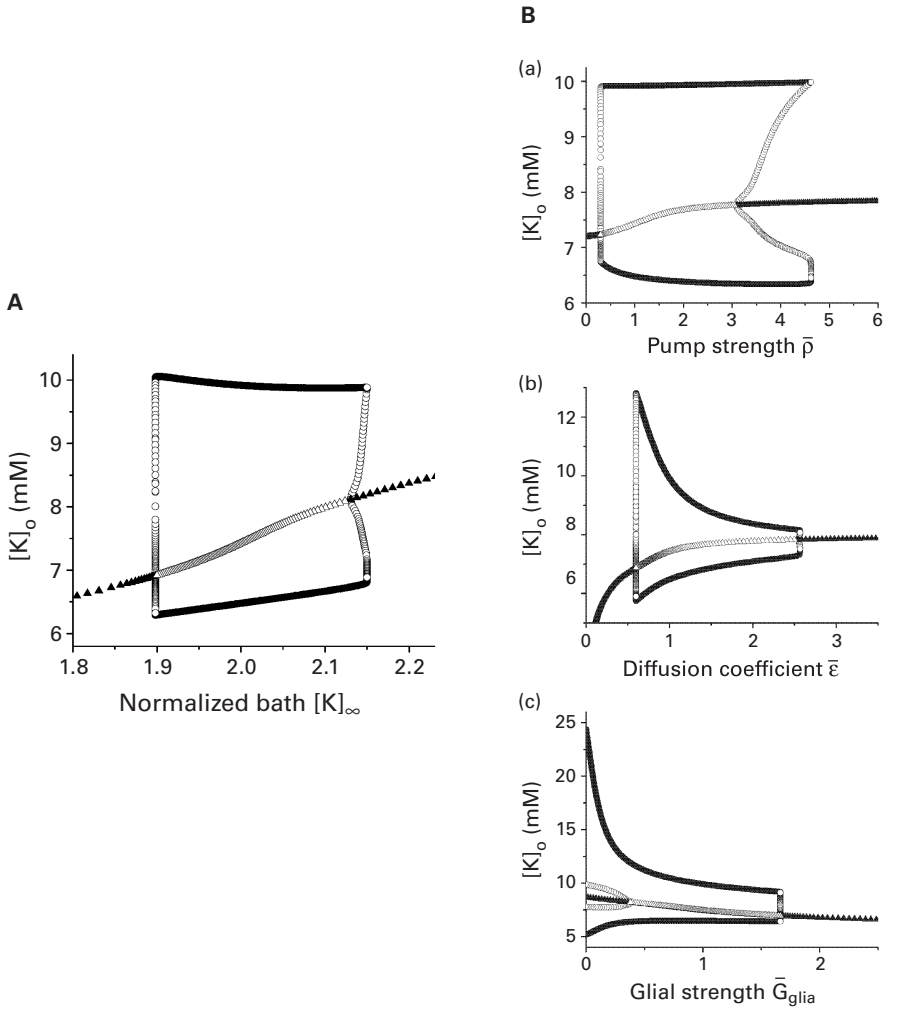


Figure 12.5
A shows the bifurcation structure of extracellular potassium, $[K^+]_o$ as a function of distant reservoir (bath or vasculature) concentrations of potassium $[K^+]_\infty$ normalized to the normal physiological bath concentration. Thus, 2.0 indicates twice normal bath concentration where oscillations typically begin in experiments [RLJ85]. B shows comparable bifurcation diagrams for $[K^+]_o$ as a function of ion pump strength, I_{pump}, diffusion strength, I_{diff}, and glial buffering strength, I_{glia}. The overbars, $\bar{\cdot}$, again indicate normalized with respect to normal values. Reproduced from [CUZ$^+$09] with permission.

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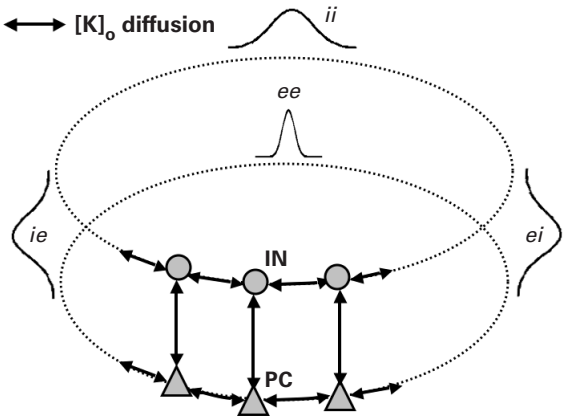


Figure 12.6
Topology of the network. The network consists of one pyramidal cell layer and one interneuron layer, both arranged in a ring (dotted lines). The two neuronal types make synaptic connections in the same layer as well as across the layers, with Gaussian synaptic footprints (represented by Gaussian curves, where ee, ei, ie, and ii stand for excitatory-excitatory, excitatory-inhibitory, inhibitory-excitatory, and inhibitory-inhibitory synaptic connections, respectively). Note the narrower ee Gaussian curve. The $[K^+]_o$ concentration around each neuron diffuses to the nearest neighbors in the same layer and the nearest neighbor in the next layer (represented by solid lines with double arrows). Reproduced from [UCJBS09] with permission.

side of figure 12.5A). In figure 12.5B, we show comparable bifurcation diagrams at a fixed value of $[K^+]_\infty$ of twice normal as a function of pump strength, diffusion, and glial buffering strength. Note that, as glial buffering strength approaches zero, the dynamics remain unstable. Oscillations abound in this quadrangle of interactions. Importantly, such oscillatory regimes are broad in parameter space. As we will shortly see, *modeling these interrelations is critical for the assimilation of data from real dynamics within the brain.*

Let's now apply our intuition from the reduced model to networks of neurons with full-model ionic dynamics. There are few parts of the brain or spinal cord where excitatory and inhibitory neurons are not tangled in an interactive web from which the dynamics of our thoughts emerge. In recent years, we have come to recognize, to our computational horror, that there are many more types of inhibitory cells than excitatory ones. Since they are small, they are hard to impale with electrodes. And they appear to orchestrate the more complex dynamics of the larger excitatory neurons [SK05]. These inhibitory neurons have unique spatial locations and particular electrical characteristics [GWM00], and they are chemically and transcriptionally unique [MTRW$^+$04].

We start with a simple framework of interacting excitatory and inhibitory neurons, as illustrated in figure 12.6.

Gutkin and colleagues [GLC$^+$01] counterintuitively demonstrated that such networks would form states of *persistent activity* by *asynchronous* interactions between its neurons (as shown in figure 12.7A). Such persistent states are a model of brief working

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memory [MBRW03, Wan03] that we feel is held electrically in the activity of neurons [FBGR89, Fus95, GR95, MED96, RAM98, RBHL99]. Recently, the phenomenon of *up* states have been similarly linked to persistent states [SVM00, SHM03]. The brain requires a certain degree of instability to form such an activity balance, often likened to a Turing instability [Mur03]. We asked, What are the network properties that render such physiological persistent states stable to perturbations and, especially, in what circumstances do such perturbations lead to seizure-like activity? In figure 12.7A, we replicate the fundamental findings of [GLC$^+$01]. Stimulating such a network can produce persistent activity that is asynchronous, and a second stimulus can synchronize and stop the activity. In order to take into account the findings of Jokubas Ziburkus and colleagues [ZCBS06] of excitatory-inhibitory cell interplay during seizure-like events, Ghanim Ullah modified the synaptic current entering the pyramidal cell and interneuron to account for the lack of spiking during depolarization blockade.[5] In figure 12.7B, we show a broad regime for given levels of $[K^+]_\infty$ within which stable persistent activity is a function of the strength of excitatory synaptic activity (α_{ee}). We then studied how such persistent activity would respond to perturbations (figure 12.7C) as a function of $[K^+]_\infty$ (figure 12.7D). We see that, as a function of $[K^+]_\infty$, the persistent state can be stable, or lose stability and involve the entire network in a seizure-like state. *Network ion dynamics may be an important aspect of the determination between normal and pathological dynamics.*

In recent years, the role of glial cells in such ion homeostasis has become increasingly recognized. Astrocytes in particular have been shown to affect virtually every aspect of neuronal function through glia-neuron crosstalk. They have the ability to buffer excitatory ions such as K^+ from the interstitial volume [AMM02], preventing an undue increase in excitation of the neuronal network. They can also enhance the excitability of the network

5. We use

$$I^e_{syn} = -\frac{(V^e_j - V_{ee})}{N}\sum_{k=1}^{N} g^{ee}_{jk} s^e_k \chi^e_{jk} - \frac{(V^e_j - V_{ie})}{N}\sum_{k=1}^{N} g^{ie}_{jk} s^i_k \chi^i_{jk}$$

and

$$I^i_{syn} = -\frac{(V^i_j - V_{ei})}{N}\sum_{k=1}^{N} g^{ei}_{jk} s^e_k \chi^e_{jk} - \frac{(V^i_j - V_{ii})}{N}\sum_{k=1}^{N} g^{ii}_{jk} s^i_k \chi^i_{jk}$$

where $V^{e/i}_j$ is the voltage of the jth excitatory/inhibitory neuron, $s^{e/i}_k$ is the variable giving the temporal evolution of the synaptic efficacy emanating from the kth excitatory/inhibitory neuron, and $\chi^{e/i}_{jk}$ takes into account the interplay between pyramidal cells and interneurons described earlier; that is, if a cell goes into depolarization block, then the synaptic inputs from that cell to others are reduced toward zero by the factor $\chi^{e/i}_{jk}$. Vee, Vei, Vie, and Vii are the reversal potentials for the excitatory-excitatory, excitatory-inhibitory, inhibitory-excitatory, and inhibitory-inhibitory synaptic inputs, respectively. This is necessary in order to be able to handle the voltage region of steady depolarization block, especially in the presynaptic cells (generally about -30 to -10 mV with respect to outside the cell). See [US09] for further details. Such a formulation leaves the synaptic effects of fast-spiking neurons (when potential overshoots zero) unchanged.

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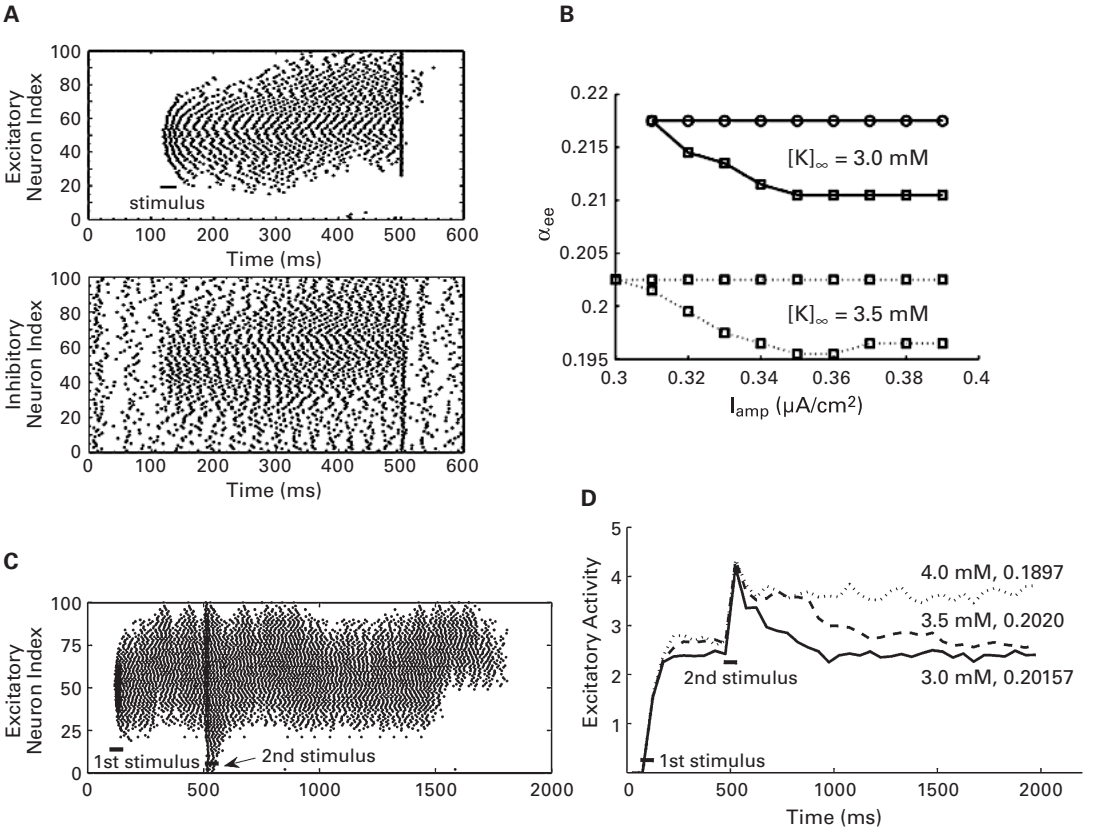


Figure 12.7
The existence and stability of spatially restricted foci of activity in neuronal network of excitatory and inhibitory neurons. In A, a 20-msec excitatory stimulus with amplitude $I_{stim} = 1.5\mu$A/cm^2 applied to pyramidal cells 21–79 causes a persistent and spatially restricted activity packet in the network. This activity is turned off by synchronizing neuronal activity using a stronger 1-msec stimulus of 100μA/cm^2 at $t = 500$ msec. The top (bottom) panel shows a raster plot for the pyramidal cell (interneuron) network where each dot represents a single-cell spike in the network. Here, $\alpha_{ee} = 0.215$, and $[K^+]_\infty = 3.0$ mM. B shows regions of persistent activity stability for different values of $[K^+]_\infty$. Beyond a certain threshold of the stimulus ($I_{stim} = 0.31$) the weakness of the stimulus is compensated by the synaptic strength and external potassium level. In C we show the results of a first stimulus at t = 112 msec that causes a stable focus of activity in the network. A second stimulus t = 500 msec causes the focus to take over the entire population. After the stimulus is removed, the network decays to zero-activity state, returns to the stable focus, or expands to the state where the activity is spread throughout the entire population. D shows the effect of different $[K^+]_\infty$ values on the first and second stimuli. For lower extracellular potassium, the network activity decays back to the stable focus after the second stimulus is turned off. However, if extracellular potassium is higher, the network maintains its enhanced activity even after the second stimulus vanishes. Reproduced from [UCJBS09] with permission.

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by releasing glutamate and ATP in a Ca^{2+}-dependent manner [PBL+94, PH00]. In spite of a direct role of glia in modulating the activity of neuronal networks, detailed models of glia-neuron interaction are few [NJ03]. In our modeling results presented earlier, we demonstrated how dysfunction of glial cells can cause the *occasional* transition from normal stable neuronal activity to seizure-like uncontrolled activity. Several experimental studies support glial dysfunction during epileptic seizures [HGJ+00, HSH+00]. In Ullah and colleagues [UCJBS09], we further modeled the effect of glutamate released from glia in a Ca^{2+}-dependent manner [TAT+05, KXX+05]. Our simulations showed that the ability of glial cells to induce seizure-like activity depends on the initial baseline state of the network. If the network is in a relatively low excitatory state, then a transient glial glutamate perturbation could not cause epileptic activity no matter how strong the perturbations were. However, even a small perturbation by glial cells is enough to cause seizure-like activity in the network if its excitability setpoint is sufficiently high.

An extensive literature describes the stability of ensemble systems [For90], and indeed the fundamental hallmark of a stable (equilibrium) system in physics is reflected in how such systems dissipate fluctuations. Might an analogous principle regarding the response to perturbations help us characterize the stability of biological systems? The earlier models show that *how neuronal networks respond to perturbations, and how this response decays (or grows), are likely critical determinants of the state of the brain*. How neuronal networks respond to perturbation determines how transient patterns of activity emerge from a background activity in response to internal fluctuations of activity or external stimuli. We propose that the response of a balanced, or mildly imbalanced, network to such perturbations is a fundamental feature that may underlie the initiation of a variety of disparate transient phenomena such as seizures, working memory, up-states, cortical oscillations [HTY+04], and spinal cord [CMO06] and hippocampal neonatal burst firing [LKC+02].

There is gathering evidence characterizing the interplay between excitatory and inhibitory neurons during seizures [FTIKT04, PVC99]. Ziburkus and colleagues [ZCBS06] observed interplay between pyramidal cells and interneurons during in vitro seizure-like events, as shown in figure 12.8A. In particular, pyramidal cells were seen to go into a silent state when interneurons were burst firing, followed by burst firing in pyramidal cells when interneurons went to depolarization block. The interneurons appeared to be firing in a much more erratic and high-frequency manner than the pyramidal cells, and a particularly large fluctuation might have tipped the interneuron network into depolarization blockade. What might be the origin of the depolarization block in such interneurons? Figure 12.8B shows that such blockade can be emulated in this model network by the effect of potassium dynamics. In particular, $[K^+]_o$ builds up in the extracellular space, sculpting the activity of the interneurons in a fast-slow-fast temporal pattern, with depolarization blockade in the middle of the episode. It is our view that such ion dynamics likely are a factor in these events, but it is hard for someone who has experience with oxygen physiology to be able to observe such depolarization blockade and not consider the contribution of oxygen limitation during high-frequency firing to this phenomenon as well [Som04, SS87].

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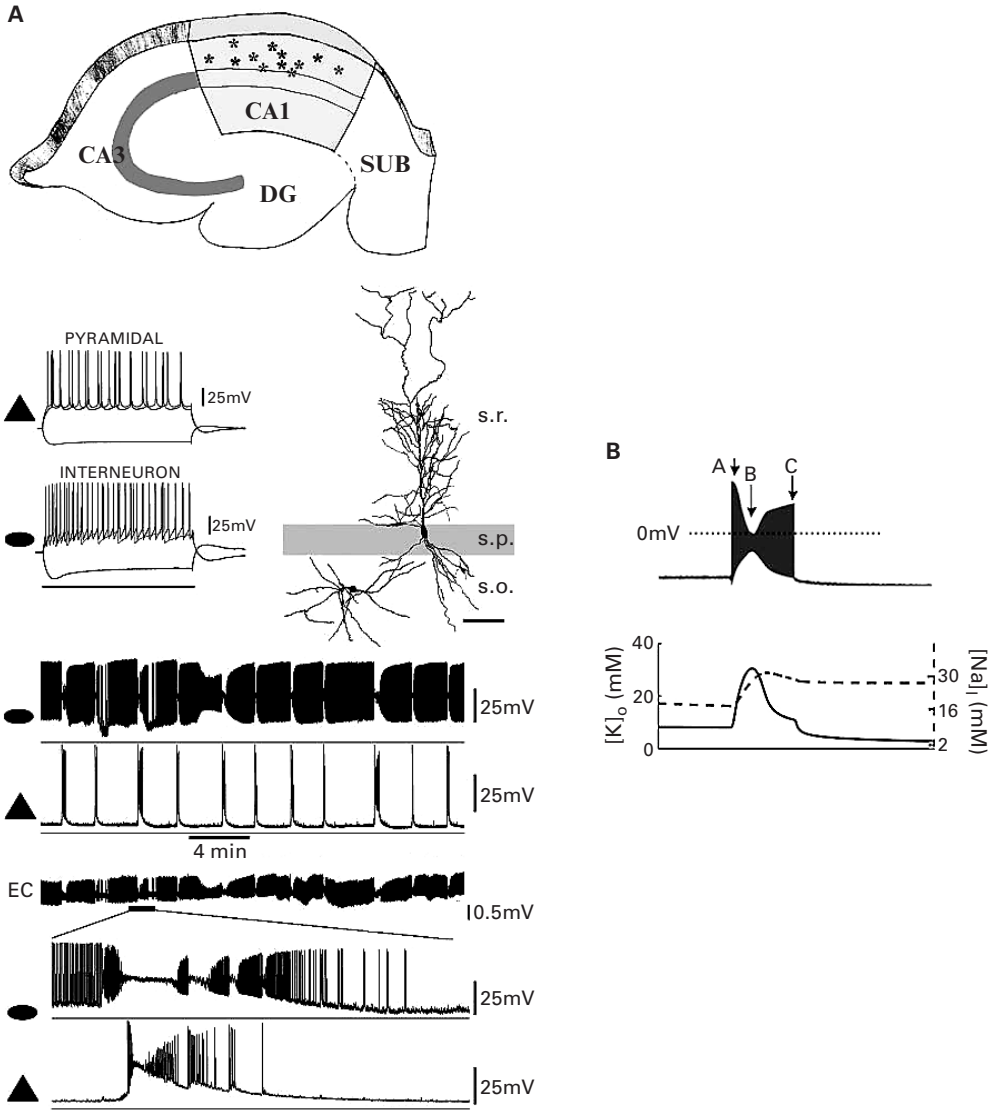


Figure 12.8
Excitatory-inhibitory interplay during seizurelike events in vitro. In A, at top, is a schematic with gray stars indicating interneurons that underwent depolarization block in seizures. Below this are examples of pyramidal (triangle) and oriens interneuron (ellipse) membrane properties following 500-msec (scale bar) negative and positive current injections. Camera lucida microscope drawings of the dendrites of the cells. s., stratum; o., oriens; p, pyramidale; r, radiatum. Scale bar: 50 μm. The lower electrical recordings show repetitive seizures and the relationship of excitatory and inhibitory cell interplay. The interneuron (ellipse) is more active than the pyramidal (triangle) cell. The extracellular (EC) electrode was placed in the oriens, about 75 μm from the interneuron soma. Note the cessation of axonal spikes in EC during interneuron depolarization block. Reproduced from [ZCBS06] with permission. B shows model-based network results for excitatory and inhibitory cell dynamics in the presence of active potassium dynamics. The sculpting of the interneuron activity (top panel) shows a period of depolarization blockade flanked by high-frequency activity. Reproduced from [UCJBS09] with permission.

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Keep in mind the pattern of excitatory and inhibitory neuron interplay shown in figure 12.8. We will return to this phenomenon shortly, and show how ion dynamics are as critical as neuronal interactions in a control framework for data assimilation.

12.4 Control of Single Cells with Hodgkin-Huxley and Potassium Dynamics

Ghanim Ullah [US09] followed the strategy of [SS08] to estimate the control vector to apply to the Hodgkin-Huxley equations. In figure 12.9A we show, on the left, the schematic of direct proportional control, where the control signal is generated from the noisy measurement y, which in turn is generated by adding noise to the observation variable. In the right panel, we show Kalman observer control where the model observer system is used to generate the control vector from the estimated observable $\hat{y}$. In both schemes, the control vector is applied to both the system and observer (contrast with figure 6.22 in chapter 6). In figure 12.9B, we demonstrate use of voltage measurements (top) to control the dynamics of the cell. The extracted Na^+-channel gating variable, m, is shown in the middle panel. As is clear from the bottom panel, the energy required for the control vector estimation through Kalman observer control is smaller than that used in direct proportional control. In fact, the energy requirements for direct proportional control relative to the energy requirements for Kalman observer control increases exponentially with increased noise in the measurements (inset right bottom), although beyond a certain limit the direct proportional controller becomes unstable (not shown).

Since we can estimate the rest of the gating variables and parameters along with $[K]_o$, it is possible to construct control signals based on the estimated gating variables. Such a control framework offers a novel means to modulate neuronal dynamics through various conductances using dynamic clamp methods [GM06, PAM04]. In figure 12.10A, we show a dynamic clamp simulation where seizure-like activity is controlled through g_k. In typical dynamic clamp methods, the current corresponding to a given ion species, I_i, is calculated in isolation from the relation $I_i = g_i(V\text{-}V_i)$, where g_i is the voltage-dependent dynamical conductance, and V_i is the reversal potential for a given ion species i [GM06, PAM04]. In the bottom panel we show $[K]_o$ estimated by the filter. Our state estimation framework has the advantage over this previous methodology in that the current to be injected into the cell to modulate a conductance can be calculated from more complete dynamical descriptions of available conductance and gating parameters, combined with an optimal strategy for handling model and measurement uncertainty, which, heightened by computational constraints, are major issues in applying dynamic clamp [BLSW08]. Wrapping dynamic clamp algorithms within our framework will increase the computational load for such applications, and a trade-off between increasing algorithmic complexity for accuracy versus lengthening the discrete time interval for computation will be required. Experimental testing of the predicted advantages of such a strategy given presently available hardware is under investigation.

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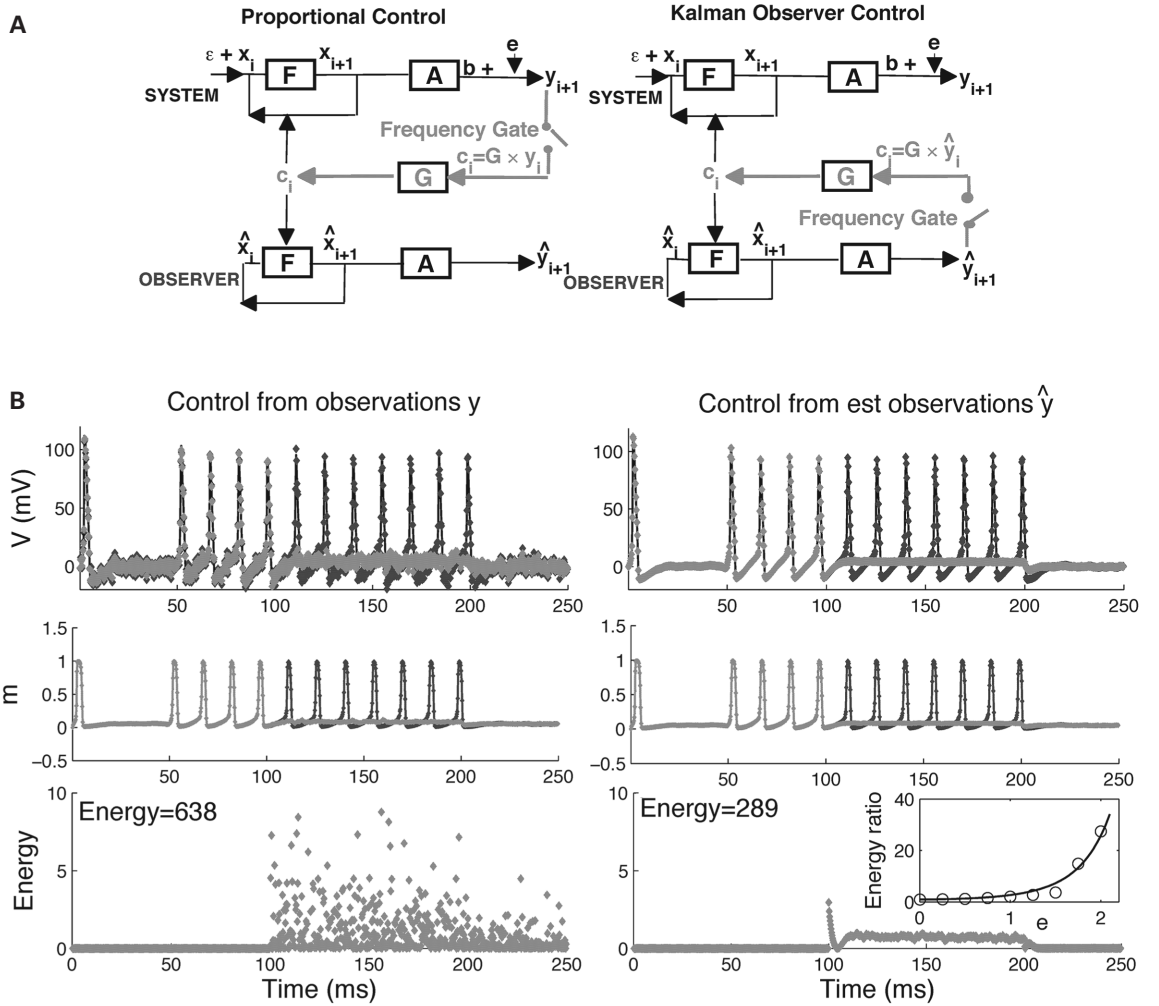


Figure 12.9
In A are the schematics of Kalman filter observers: The system function F acts on state variable x, and the observation function A produces output vector b. Output observable y is generated by adding measurement noise e to b. In the left panel of A, the control signal is constructed directly from actual noisy data by multiplying it with gain G; in the right panel, the filter uses the estimated state from the observer. B shows the results of controlling the dynamics of a Hodgkin-Huxley neuron (controlled estimates are light gray; uncontrolled estimates and true values are black) using direct proportional (left) and Kalman observer control (right). The panels from the top to bottom show V, m, and energy used by the controller (sum of squares of control vector). The control vector c is turned on at $t = 100$ msec and is only injected into V. Inset shows the ratio of energy required in direct proportion to the energy required in Kalman observer control, versus measurement noise (circles), fitted by function $\exp(e^{1.7})$. Reproduced from [US09] with permission.

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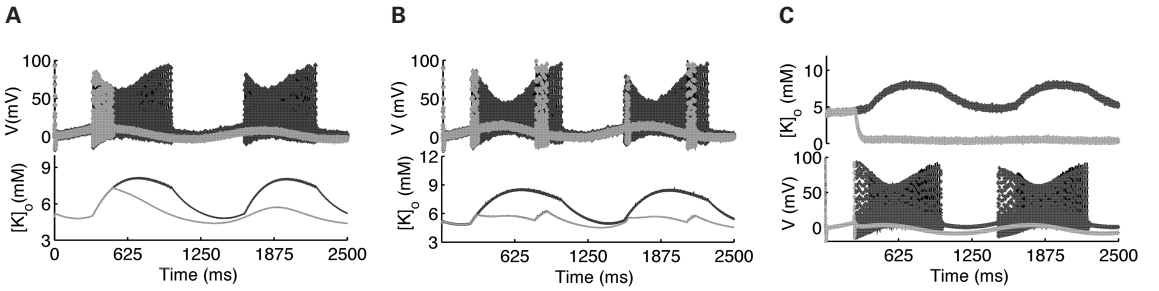


Figure 12.10
Single-cell seizures controlled through g_K (A), V (B), and $[K]_o$ (C) In A, the top panel shows V measurements with (light gray) and without (black) control. At $t = 500$ msec, g_K is doubled using dynamic conductance clamp, which suppresses the seizure-like events. Bottom panel shows estimated $[K]_o$. In B we repeat simulation from A, but the control signal is applied to V when the spiking frequency rises above 50 Hz. $[K]_o$ is lower after seizures are blocked (bottom panel). In C, neuronal dynamics including V are generated from, and control signals are applied to, $[K]_o$ measurements. The control vector is injected only into $[K]_o$ (top panel). After the injection of control signal V stays close to the resting value (bottom panel). Reproduced from [US09] with permission.

In figure 12.10B, we use V measurements to control seizure-like events by applying the control signal to V (top). Here we place a frequency gate on the filter (figure 12.9A), tuning the filter such that it applies the control signal only when the frequency of measured or estimated neuronal spiking goes above a certain value (50 Hz). The frequency gate in the filter allows the controller to modulate only the high-frequency spiking attributed to seizure-like behavior, while allowing the neurons to perform their more normal (low-frequency) functions. This not only minimizes the energy requirements by shutting the controller off when it is not needed, but it can also minimize the long-term risk of damage to brain tissue due to continuous electrical stimulation.

An interesting variant using our framework is to measure $[K^+]_o$ instead of voltage. This is a slower signal, but one that sets and reflects the excitability of the network. The relatively small extracellular space causes a small I_K to generate moderate changes in $[K]_o$, which enhances the membrane excitability significantly by changing the reversal potentials for K^+ channels. We therefore reconstruct neuronal dynamics including V based on $[K]_o$ measurements. In the top panel of figure 12.10C, we show $[K]_o$ measurements, and apply K^+ control at $t = 250$ msec. Driving $[K]_o$ to lower levels causes the seizure-like burst firing in V to go to silence (bottom). Controlling seizures with $[K]_o$ has two advantages: (1) the time scale for $[K]_o$ is slower than the fast dynamics of V and gating variables, and hence can be more easily implemented in real time, and (2) the $[K]_o$ controller allows us to control the neuronal network rather than a single cell. It has been shown, using ion-selective microelectrodes to measure K^+ activity changes in extracellular space beneath the surface of the brain, that electrical current could be used to decrease or increase $[K]_o$ [GMN83]. A recent theoretical study [BMS08] predicts a

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functional block along axons due to K^+ accumulation in the extracellular (submyelinated) space during deep brain stimulation for Parkinson's disease. Recent optical techniques offer the prospect of increasing the speed and decreasing the scale of $[K]_o$ measurements [MFL89].

12.5 Assimilating Seizures

The purpose of this long chapter was to get to the point of utilizing all of the preceding in an experimental situation. In the experiments shown in figure 12.8A, we recorded from two or three neurons simultaneously, including at least one inhibitory and one excitatory neuron. Such data offers us a way to employ our network models and *validate* our reconstruction through data assimilation. We can use the data from one excitatory (inhibitory) cell and reconstruct the other inhibitory (excitatory) cell. Can we achieve Jule Charney's [Cha51] description of "first approximations that bore a recognizable resemblance to the actual motions" [Cha51]?

In Ullah and Schiff [US10], we created a network model of two cells, one inhibitory and one excitatory, as shown in figure 12.11. Using this model, we can take recordings from either pyramidal inhibitory cells, and reconstruct the remaining unobserved variables in the system as shown in figure 12.12.

Including the dynamics of intra- and extracellular ion concentrations in the network model is *necessary* for accurate tracking of seizures. Using Hodgkin-Huxley–type ionic currents with fixed intra- and extracellular ion concentration of K^+ and Na^+ ions fails to track seizure dynamics in pyramidal cells, as shown in figure 12.12Bc. We used physiologically normal concentrations of 4 mM and 18 mM for extracellular K^+ and intracellular Na^+, respectively, for these simulations. The conclusion remains the same when higher $[K]_o$ and $[Na]_i$ are used. A similar tracking failure is found while tracking the dynamics of inhibitory interneurons during seizures (data not shown).

Reconstructing Network Interaction

Since the interaction of neurons determines network patterns of activity, it is within such interactions that we seek unifying principles for epilepsy. To demonstrate that an unscented Kalman filter framework can be used to study cellular interactions, we reconstructed the dynamics of one cell type by assimilating the measured data from another cell type in the network in [US10]. In figure 12.13, we show the estimated membrane potentials, but we also reconstructed the remaining variables and parameters of both cells (not shown). We first assimilated the membrane potential of the pyramidal cell to estimate the dynamics of the same cell and also the dynamics of a coupled interneuon in figure 12.13A–D. Conversely, we estimate the dynamics of pyramidal cell from the simultaneously measured membrane potential measurements of the interneuron (figure 12.13E–F). As is evident from

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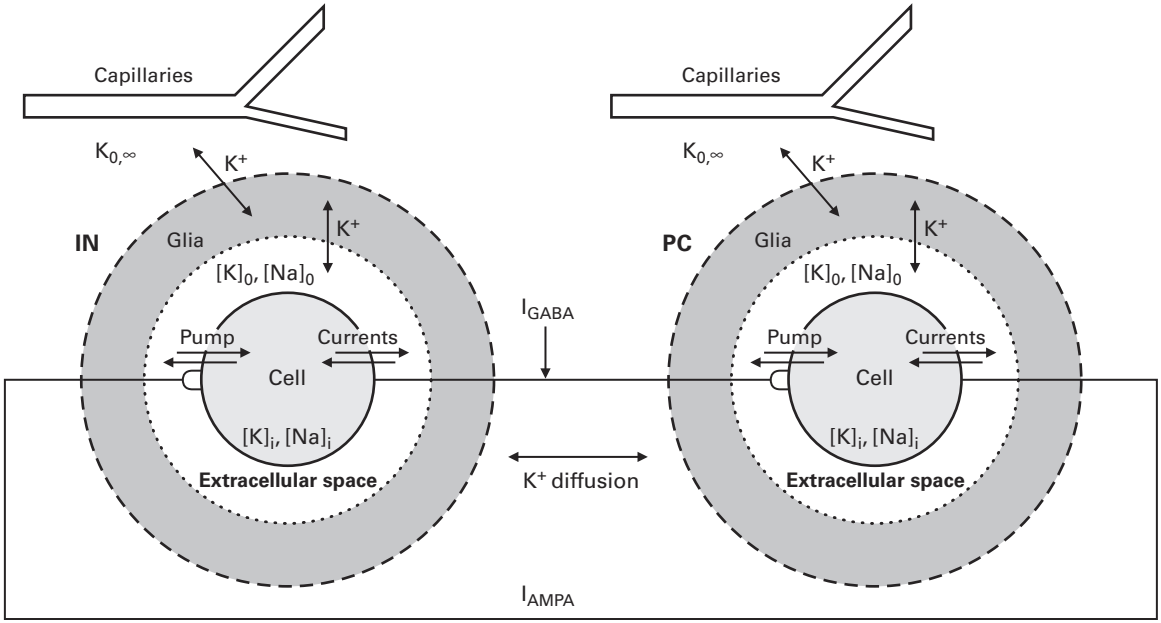


Figure 12.11
A schematic of the model dynamics: Potassium is released to the extracellular space and is pumped back to the cell by the ATP-dependent K^+-Na^+ exchange pump, buffered by glia, and diffuses to the microenvironment where it interacts with capillaries. Sodium entering the cell through Na^+ channels is pumped out of the cell by the ATP-dependent pump. Pyramidal cell (PC) and interneuron (IN) from the CA1 region of the hippocampus are coupled both synaptically and through lateral K^+ diffusion. Figure courtesy of G. Ullah.

figure 12.13, the Kalman filter framework is successful at reciprocally reconstructing and tracking the dynamics of these different cells within this network.

It appears that we have met Charney's challenge [Cha51].

12.6 Assimilation in the Intact Brain

I view the reconstruction of individual cellular activity from multiple intracellular impalements in brain slice experiments as proof of concept. A bit of the neuroscientist's equivalent of a party trick. We are presently working with an even harder trick: the assimilation of three types of neuron from such experiments—oriens interneurons, basket cells, and pyramidal cells. These appear to be the minimum three canonical cell types to even begin to consider that we may be able to capture essential dynamics in small bits of the brain. Much of the nervous system has small inhibitory cells that lay near the principal excitatory output cells. These cells receive excitation from branches of the excitatory cells, and deliver inhibition back near the cell bodies of those excitatory cells. Such inhibition near the perisomatic sites

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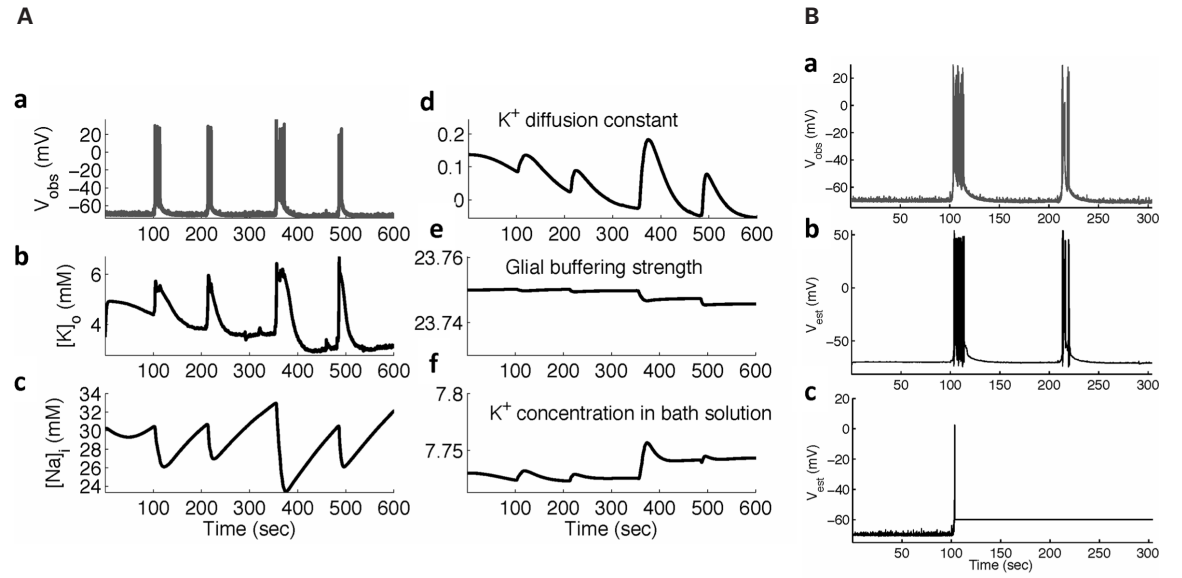


Figure 12.12
Assimilation of spontaneous seizure data by whole-cell recording from pyramidal neurons. Aa shows measured V (gray) from a single pyramidal cell during spontaneous seizures. Estimated (black) $[K]_o$ (Ab), $[Na]_i$ (Ac), K^+ diffusion constant (Ad), glial buffering strength (Ae), and K^+ concentration in bath solution (Af). Data generously provided by Jokubas Ziburkus; panel Aa modified from [ZCBS06] with permission. B shows an attempt to assimilate such data without including microenvironmental K^+ and Na^+ dynamics in the model. Observed (Ba) and estimated (Bb) membrane potential using the model with ion concentrations dynamics. In Bc, we show estimated membrane potential using the model without ion concentrations dynamics. Reproduced from [US10] with permission.

where action potentials are initiated are well positioned to cut off the outputs of those cells. In the hippocampus, these cells have a high level of calcium-binding protein parvalbumin within them, which helps in their identification [SK05]. In cortex we tend to call these "rapid spiking" cells. In the spinal cord, the Renshaw cell takes this role. Our nervous systems are full of cells that act as speed governors of our large neurons—they keep them in line. Perhaps given recent data, they also play an important role in forming patterns of activity.

There is also another type of inhibitory cell with wide distribution. We focused on these in the preceding work. In the hippocampus, about 80% of the neurons in the *stratum oriens* layer of the hippocampus are known as oriens-lacunosum-moleculare interneurons—a dreadful name that derives from the course of their axons, from the oriens layer to the border of the lacunosum-moleculare layer. In the cortex, these are known as low-threshold spiking interneurons [GBC99]. The importance of this is that the output of these inhibitory cells, after being activated by the pyramidal cells, bypasses the pyramidal cell bodies and courses all the way to the tips of the principal (apical) dendrites of these pyramidal cells. This places the inhibition of these neurons in a position to cut off the inputs to these cells.

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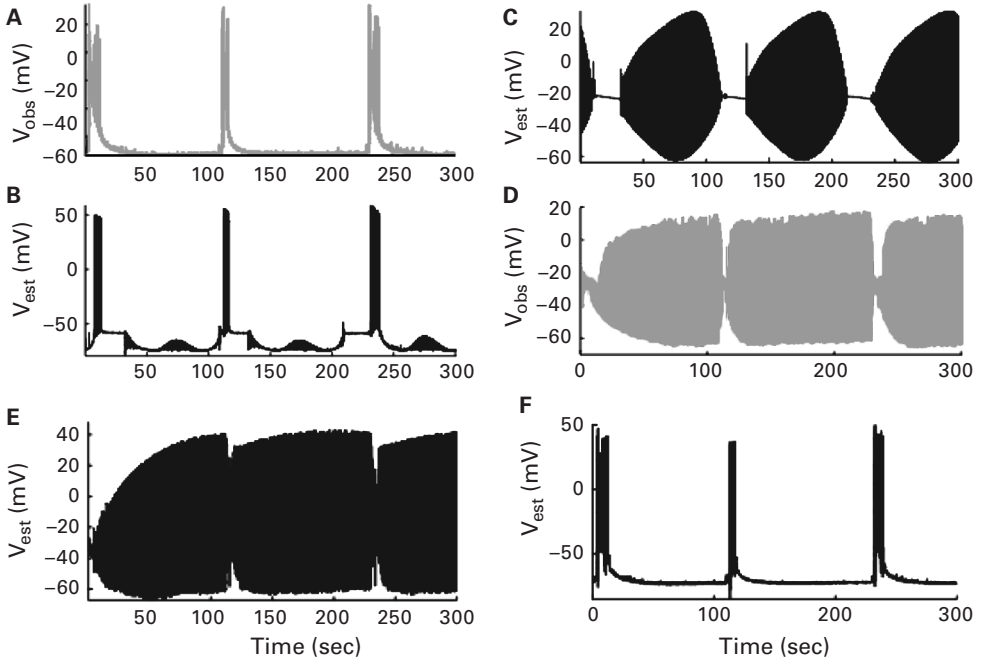


Figure 12.13
Validation of reconstruction in simultaneously recorded cells. Measured (shown in A, gray) and estimated (B, black) V for pyramidal cell. C shows estimated V for interneuron. We used the membrane potential recorded from the pyramidal cell (A, gray) to not only reconstruct the full dynamics of the same pyramidal cell (only membrane potential is shown in B) but also reconstructed the dynamics of the interneuron (only membrane potential shown in C). Simultaneously recorded V from the interneuron is shown in (D, gray) for comparison. Estimated V for interneuron in E and pyramidal cell in F by assimilating measured V from interneuon shown in D. D through F are converses of A through C. That is, in D through F we used membrane potential recorded from the interneuron (shown in D) to not only reconstruct the full dynamics of the same interneuron (only membrane potential shown in E) but also the coupled pyramidal cell (only membrane potential shown in F; compare with actual values shown in A). Simultaneous membrane potential measurements, shown in (A and D), were from a pyramidal cell and oriens interneuron in the hippocampus using simultaneous dual whole-cell patch clamp recordings demonstrating the firing interplay between these cells during in vitro seizures. Data provided by Jokubas Ziburkus. Panels A and D are modified from [ZCBS06] with permission. Overall figure reproduced from [US10] with permission.

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Gibson and colleagues [GBC99] discovered that these different types of inhibitory interneurons in cortex tend to electrically couple with their own cell types and act as a larger subnetwork in their synchronized activity. It was further discovered by Gloveli and colleagues [GDR+05] that these subnetworks can, at least in the hippocampus, be spatially orthogonal to each other. Tort and colleagues proposed [TRD+07] a way to link the computational models of Gloveli and colleagues [GDR+05] together to reflect the binding of our hippocampal slices back together into their original functional arrangement.

We have taken the model from Gloveli and colleagues [GDR+05] and simplified it somewhat for data assimilation, as shown schematically in figure 12.14. We employ our three canonical neurons—pyramidal, basket, and oriens—and arrange their synaptic connections in the native way they interconnect. We then add the potassium dynamics as before. The schematic shows an extracellular recording, since this is the customary arrangement that we use in the awake behaving brain [SCP+07, SCP+09]. The modeling must emulate an appropriate observing function reflecting the resistive and capacitive properties of the brain lattice within which these neurons are embedded, and helping to best process the extracellular signal.

We use our intracellular experiments, as before, to set many of the unknown parameters of this observer model. Ideally, we will track only the parameters that are most essential to track in such scenarios. The schematic then shows the assimilation of the estimated potential of the lumped dendrite compartment or equivalent cylinder from such data recording. We then reconstruct the potential of the pyramidal soma (E), basket cell (I), and oriens cell (O). We are presently employing such model-based data assimilation in several in vivo paradigms: spontaneous chronic seizure activity, and navigational oscillatory rhythms in maze settings. The remainder of this figure, showing the results of this effort, will gradually materialize in the years to come.

12.7 Perspective

To our knowledge, these are the first demonstrations of data assimilation from neuronal networks using fundamental model-based techniques. It is important to frame out this statement. There has been intense interest in the neuroscience communities in bringing control-theoretical tools to bear on neuronal encoding and decoding problems as discussed in chapter 9 [BNF+01, BWFB05, SB03, SB07, SEMB07, SWSB07, WGB+06, EBE+07, LOH+09]. In all of this work, statistical models (continuous or point process) were fit to data recorded from neurons, and these empirical models incorporated into applications. Our use of control theoretic tools is very different. We built computational models from the physiological properties of neurons and their networks, as well as the properties of ion metabolism, *without data fitting*. Using these fundamental models of the physics of neuronal systems, we fuse these models with data—*data assimilation*—in a manner commonly applied in

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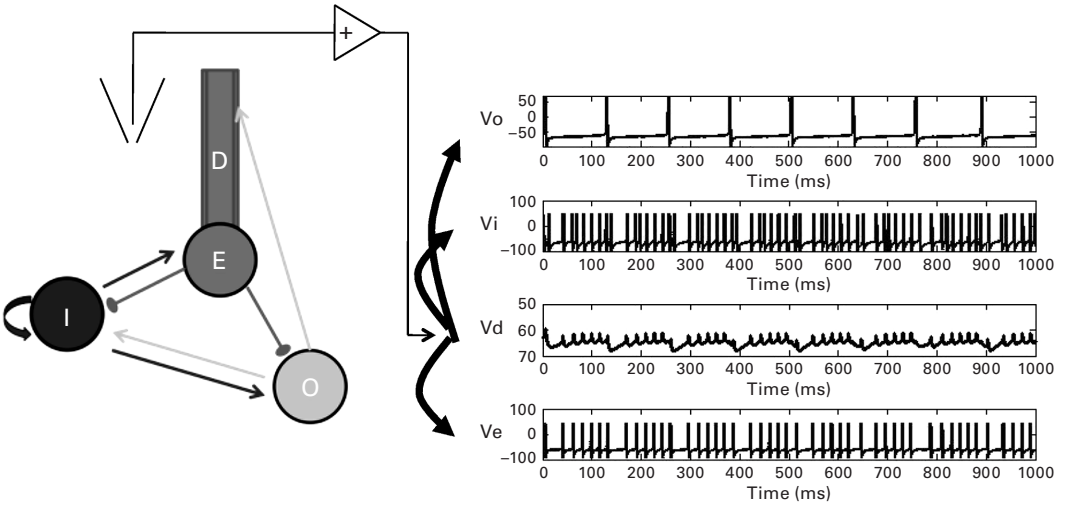


Figure 12.14
Model of three canonical neurons interacting within a transverse "slice" of the hippocampus: pyramidal cells (soma, E, and dendrite, D), basket cells (I), and oriens interneurons (O). We envision the recording of extracellular voltage with the electrode and amplifier shown, and through appropriate filtering assimilate this potential into the dendrite voltage, V_d. One could similarly record and assimilate the pyramidal soma potential. In the schematic, we use arrows to represent the reconstruction through the model of the potential within the oriens neurons, V_o, basket cells, V_i, and pyramidal soma potential, V_e. Figure courtesy of Yina Wei.

meteorology [Kal03, HKS07, OHS$^+$04, BHK$^+$06, SBIJ08, SKJI06]. Other authors have also recently discussed the importance and power of going beyond statistical empirical models in neuronal systems, and simulations have begun to explore the feasibility of carrying this out [PAF$^+$, HP09]. To our knowledge, our findings are the first experimental validation that a fundamental biophysical model of part of the brain can be employed to assimilate incomplete data and reconstruct its network dynamics.

Brain dynamics emerge from within a system of apparently unique complexity among the natural systems we observe. Even as multivariable sensing technology steadily improves, the near infinite dimensionality of the complex spatial extent of brain networks will require reconstruction through modeling. Since, at present, our technical capabilities restrict us to only one or two variables at a small number of sites (such as voltage or calcium), computational models become the *lens* through which we must consider viewing all brain measurements [MB08]. We need reconstruction to account for unmeasured parts of the neuronal system, relating microdomain metabolic processes to cellular excitability. As we learn how to do this, we can validate our cellular dynamical reconstruction against actual measurements.

At the end of this book, we have come to the beginning. We have constructed a framework of model-based data assimilation and control for spiking neuronal systems based on the

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foundational Hodgkin-Huxley ionic model. This framework has a wide range of potential applications to various systems in biology, where one can use a single measured variable to estimate the variables and parameters that are experimentally inaccessible, and allows us to predict the future trajectory of the system. These methods are applicable to a variety of model-based scenarios not discussed in this book. Recent developments in the modeling of blood pressure [ZRC07] make this a fruitful area for study. Other systems with sophisticated models and sensing available include the spiking dynamics of heart cells, or the insulin-secreting cells in the pancreas and blood glucose regulation.

Nevertheless, in terms of brain dynamics, I would wish to suggest to the reader that as our models of the nervous system improve, we should strive to never again settle for inspecting raw data. Our microscope of understanding will consist of filtering through our biophysical models what we measure and observe from the brain.

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